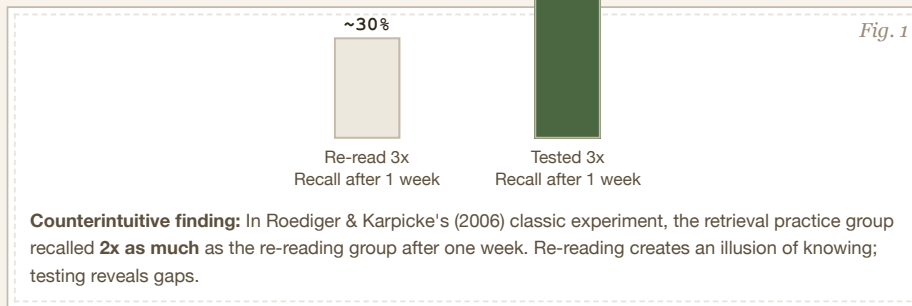


Retrieval Practice: The Power of Active Recall

提取练习：主动回忆的力量

Why testing yourself beats re-reading · Printable Plate

I. THE TESTING EFFECT · THE TESTING EFFECT



II. OBSERVATIONS · OBSERVATIONS

- Fig. 2
- Obs. 1 **"Desirable difficulties" hypothesis** — Bjork (1994) found: the more effortful retrieval is, the stronger the memory trace. Easy learning (highlighting) feels good but works poorly; difficult learning (closed-book recall) feels bad but works well.
 - Obs. 2 **Feedback determines ceiling** — Retrieval practice must include feedback. Knowing right/wrong immediately after testing converts wrong answers into correct memories. Testing without feedback may reinforce errors.
 - Obs. 3 **Spacing + Retrieval = strongest combo** — Spaced repetition or retrieval practice alone are effective, but combined they are optimal (Carpenter et al., 2012). Retrieving at spaced intervals is the most effective known learning strategy.
 - Obs. 4 **Testing is not just assessment — it's learning** — Traditional "teach → test" treats testing as the endpoint. Cognitive science proves: testing itself is the most powerful learning event. Every recall re-encodes memory, making it more stable and accessible.

III. METHODS · PRACTICAL METHODS

- Fig. 3
- 1 **Closed-book note-taking** — After reading a chapter, close the book and write everything you can recall on a blank sheet. Then check the book and fill gaps in a different color. The "gap" is what you need to strengthen.
 - 2 **Question cards (not answer cards)** — Traditional flashcards show the answer first, inducing "recognition" not "recall." Correct approach: question on front, hint/answer on back. Seeing the question forces active retrieval.
 - 3 **"Test-before-study" pattern** — Take a pre-test before formal learning. Even if you get everything wrong, pre-testing improves subsequent learning efficiency by 10-20% (Richland et al., 2009). This is the "pretesting effect of retrieval practice."
 - 4 **Feynman Technique (teach someone)** — Explain a concept in the simplest language to an imaginary 12-year-old. Where you get stuck signals shallow understanding. Far more effective than re-reading — teaching forces retrieval and organization.
 - 5 **Interleaved practice** — Don't do 20 problems of the same type at once. Mix types: 3 integrals + 2 derivatives + 4 probability. Each retrieval requires first "selecting a strategy," strengthening discrimination and transfer.

* Retrieval practice was rated by Dunlosky et al. (2013) as the learning technique with the strongest evidence base, with "high utility" recommendation.

Related Books

Make It Stick Brown, Roediger & McDaniel | retrievalpractice.org by Pooja Agarwal

memory considerations in the training of human beings. | Dunlosky et al. (2013). Improving Students' Learning. *Psychological Journal Psychology Review*, 24, 369-378.

